

# NetraJyoti AI

## Project Delivery, Ethical Considerations and Lean Canvas

This supporting capstone document explains how NetraJyoti AI can be delivered responsibly: why the product is viable, how the team will work, how work will be tracked in Jira, and how safety, privacy, and inclusion are protected.

**Product boundary.** NetraJyoti gives Bengali-first eye-care guidance and service navigation. It helps users understand safe next steps; it does not diagnose disease or replace a clinician.

| Document item                 | Value                                                                                                     |
|-------------------------------|-----------------------------------------------------------------------------------------------------------|
| Product                       | NetraJyoti AI MVP                                                                                         |
| Recommended delivery approach | Lean UX + Agile Kanban                                                                                    |
| Planning horizon              | 12-week capstone sample plan                                                                              |
| Work tracking                 | Jira project NET; Epic NET-1 and associated stories                                                       |
| Primary users                 | Rural Bengali-speaking eye-care seekers, caregivers, ASHA/community-health workers, outreach coordinators |
| Ethical priority              | Safety-first, non-diagnostic, privacy-preserving care navigation                                          |

### 1. How the deliverables fit together

The five requested items are one delivery story, not separate exercises. The Lean Canvas explains the product opportunity; ethical considerations define what the product may and may not do; the methodology explains how uncertainty will be reduced; the project plan turns that approach into milestones and ownership; and the Jira board provides visible execution tracking.

| Deliverable            | Question answered                                                 | Output                                       |
|------------------------|-------------------------------------------------------------------|----------------------------------------------|
| Lean Canvas            | Why does this product matter and how could it create value?       | Nine-block product and viability model.      |
| Ethical considerations | What safety, privacy, inclusion, and accountability limits apply? | Safeguards and governance checks.            |
| Methodology            | How will the team learn and deliver safely?                       | Lean UX discovery plus Kanban delivery flow. |

| Deliverable         | Question answered                               | Output                                                                   |
|---------------------|-------------------------------------------------|--------------------------------------------------------------------------|
| Sample project plan | When will each activity happen and who owns it? | Sequenced 12-week plan with milestones.                                  |
| Jira board          | How is work made visible and managed?           | Backlog/To Do/In Progress/Testing/Done workflow and evidence screenshot. |

## 2. Lean Canvas

This uses the default Lean Canvas structure. Assumptions should be tested through real user interviews, usability sessions, clinical/content review, and a small monitored pilot.

| Lean Canvas block         | NetraJyoti application                                                                                                                                                                                                                    | Assumption to validate                                                                                                                    |
|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| Problem                   | People in rural West Bengal may delay eye care because they do not know whether symptoms are serious, which information to trust, or where to go. Cost, travel, waiting time, low digital confidence, and language barriers add friction. | Existing alternatives: family advice, informal care, forwarded messages, generic web search, and direct travel without service certainty. |
| Customer segments         | Primary: Bengali-speaking rural users with an eye concern. Secondary: caregivers who help make decisions. Tertiary: ASHA/community-health workers and outreach coordinators.                                                              | Early adopters: users/caregivers who have a phone, a current eye concern, and uncertainty about the right care route.                     |
| Unique value proposition  | “Bengali-first safe next-step guidance for eye problems.”<br>NetraJyoti reduces confusion before treatment by turning fixed symptom inputs into clear, non-diagnostic care navigation.                                                    | High-level concept: a safety-first Bengali voice/text guide to the appropriate eye-care service.                                          |
| Solution                  | Fixed symptom selection; reviewed urgent-warning check; deterministic urgent/routine/human-support routing; district/town service direction; family sharing; optional controlled AI caregiver summary.                                    | MVP deliberately excludes diagnosis and treatment advice.                                                                                 |
| Channels                  | Eye camps, vision centres, ASHA/community-health workers, NGO/eye-care partners, posters/QR codes, local WhatsApp sharing through caregivers, and pilot referrals.                                                                        | Start with partner-supported community outreach, then assess digital referral channels.                                                   |
| Revenue streams / funding | For a social-impact pilot: NGO/CSR sponsorship, eye-care partner programme funding, public-health grant support, or service-provider implementation agreements.                                                                           | No user-facing payment is assumed in the MVP.                                                                                             |
| Cost structure            | Product design/research, clinical and Bengali-content review, frontend/backend development, cloud/database hosting, AI usage when enabled, service-directory verification, usability testing, and outreach operations.                    | The controlled-AI feature must remain optional to contain both cost and safety risk.                                                      |
| Key metrics               | Journey completion; route distribution; time to reach recommended action; service-direction use; caregiver share/copy rate; comprehension/usability score; AI fallback rate; verified-service data freshness.                             | Never use clinical outcome claims without a validated study design.                                                                       |
| Unfair advantage          | Trusted local eye-care and community-health partnerships, reviewed Bengali content, curated service data, and a safety-first workflow tailored to the rural West Bengal context.                                                          | The advantage must be maintained through regular service-data verification and user research.                                             |

### 3. Recommended methodology: Lean UX + Agile Kanban

Use a combined approach: Lean UX for problem discovery and evidence gathering, followed by Agile Kanban for visible, continuously prioritised delivery. This is appropriate because NetraJyoti has both product uncertainty (language, comprehension, service-navigation behaviour) and safety constraints that require short, reviewable increments.

| Practice              | How it is applied to NetraJyoti                                                                                                                                         | Why it fits                                                                                                 |
|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|
| Lean UX discovery     | Interview users, caregivers, ASHA workers, and outreach coordinators; map current care-seeking journey; test wireframes in Bengali; define hypotheses before building.  | Prevents building a text-heavy or clinically unsafe workflow without evidence.                              |
| Kanban delivery       | A prioritised backlog and visual workflow (To Do, In Progress, Testing / Content Review, Done), with regular review, improvement actions, and release-ready increments. | Makes work visible, limits work in progress, and enables continuous prioritisation and safe release checks. |
| Safety/content review | Review symptom labels, urgent criteria, Bengali wording, service-data disclaimers, and AI fallback before release.                                                      | Health content requires defined approval gates rather than only engineering sign-off.                       |
| Usability testing     | Run short task-based sessions on Android-sized devices, slow networks, and voice/location denied states.                                                                | Validates accessibility and practical completion, not just feature delivery.                                |
| Pilot learning loop   | Measure outcomes, collect feedback, fix high-risk friction, and revalidate content/service data before expansion.                                                       | Supports evidence-led iteration and avoids overclaiming MVP impact.                                         |

#### 3.1 Example two-week delivery cadence

| Day | Activity                             | Evidence / output                                                |
|-----|--------------------------------------|------------------------------------------------------------------|
| 1   | Planning and safety/content review   | Prioritised Jira stories, acceptance criteria, and risk check.   |
| 2–8 | Design, build, and test              | Working increment; unit/component tests; reviewed copy.          |
| 9   | Internal demo and QA                 | Defect list, route regression result, service-data confirmation. |
| 10  | Flow review and improvement planning | Stakeholder feedback, updated backlog, and improvement actions.  |

## 4. Sample 12-week project plan

This is a sample capstone plan. The dates can be replaced with your actual submission calendar. Each phase has a clear decision or artefact so that progress is measurable.

| Week | Focus                                     | Key activities                                                                    | Owner                      | Milestone                                  |
|------|-------------------------------------------|-----------------------------------------------------------------------------------|----------------------------|--------------------------------------------|
| 1    | Discovery and evidence                    | Interview/research plan; current journey; problem validation; safety assumptions. | Product owner + researcher | Problem evidence and risk log approved.    |
| 2    | Personas and Lean Canvas                  | Personas, jobs-to-be-done, Lean Canvas, success metrics.                          | Product owner              | Target users and value proposition agreed. |
| 3    | User journey and information architecture | End-to-end branches, content inventory, wireframe requirements.                   | Product + UX               | Urgent/routine/human flow reviewed.        |
| 4    | Wireframes and usability prototype        | Bengali-first prototype; test script; user feedback.                              | UX + product               | Usability findings prioritised.            |
| 5    | Technical design and backlog              | HLD, Technical PRD, API/data design, Jira stories and estimates.                  | Tech lead + product        | MVP backlog ready.                         |
| 6    | Core intake and routing                   | Consent, symptoms, urgent-sign questions, deterministic route API.                | Frontend + backend         | Core safe-route tests pass.                |
| 7    | Urgent and routine navigation             | 5A route, 5B–7 routine route, service-direction demo data.                        | Frontend + backend         | No-dead-end journey demo.                  |
| 8    | Support, sharing, and accessibility       | 5C route, family sharing, patient-detail consent, responsive UX.                  | Frontend + QA              | Share/fallback/device tests pass.          |
| 9    | Controlled AI and resilience              | Feature flag, backend-only OpenAI integration, validator and fallback.            | Backend + safety reviewer  | AI-off/error fallback confirmed.           |
| 10   | Pilot readiness and content governance    | Service-data verification, privacy review, analytics plan, release checklist.     | Outreach + product + QA    | Pilot gate decision.                       |
| 11   | Pilot/usability validation                | Task-based sessions; collect completion and comprehension evidence.               | Research + QA              | Findings and prioritised fixes.            |
| 12   | Final refinement and capstone pack        | Fix priority issues; final PRD/HLD/Jira screenshot/demo/report.                   | Whole team                 | Submission-ready package.                  |

**Dependency note.** Work on live service data, public pilot, and expanded AI capability must not begin until clinical/content review, privacy review, and operational ownership are confirmed.

### 5. Jira Kanban board, Epic and dashboard evidence

Use Jira project NET and the NetraJyoti epic/story structure to demonstrate Agile Kanban delivery. The board should reflect actual delivery state, not merely a list of issues, using its configured workflow columns.

| Board element        | Recommended configuration                                                                                                                    |
|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| Board name           | NetraJyoti AI MVP Delivery Board                                                                                                             |
| Source               | NET project; filter for the NetraJyoti epic and its child stories.                                                                           |
| Columns              | Backlog → To Do → In Progress → Testing / Content Review → Done.                                                                             |
| Swimlanes            | Optional: urgent care, routine care, human support, platform/content.                                                                        |
| Card fields          | Issue key, summary, assignee, priority, story points, and epic link.                                                                         |
| Useful quick filters | Urgent path; routine path; human support; high priority; blocked; content/service-data updates.                                              |
| Dashboard evidence   | Add a Jira listing or Filter Results gadget for the NetraJyoti epic, plus flow/WIP, status-distribution, and priority views where available. |

#### 5.1 Jira delivery links

Jira Epic: [NET-1 - NetraJyoti Epic MVP](#)

Jira dashboard: [NetraJyoti dashboard](#)

Jira Kanban board: [Board 68](#)

## 6. Ethical considerations and safeguards

NetraJyoti operates in a health-adjacent setting where unsafe communication can cause delay or false confidence. Ethical design must therefore be visible in requirements, user interface, content review, technical architecture, and operations.

| Ethical consideration                   | Risk                                                                                                                                  | Required safeguard                                                                                                                   |
|-----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| Non-diagnosis and informed expectations | The app may help users decide a safe next step but must not claim to identify a disease, prescribe treatment, or replace a clinician. | Use clear value-led consent; keep disease labels and medication advice out of content and AI output.                                 |
| Urgency and harm prevention             | Under-triage could delay care; over-triage can create fear and burden services.                                                       | Use clinician-reviewed deterministic rules, version control, tests for warning-sign combinations, and conservative urgent wording.   |
| AI safety and accountability            | Generative output can invent clinical advice, service details, or urgency.                                                            | AI is optional, backend-only, routine-route-only, feature-flagged, schema/policy validated, and always has approved fallback.        |
| Privacy and consent                     | Names, addresses, phone numbers, voice, and location are sensitive.                                                                   | Collect only optional message fields; require explicit share consent; keep data client-side in MVP; do not log/store PII by default. |
| Location consent                        | Precise location can reveal sensitive context and must never be assumed.                                                              | Ask only after a clear user action; provide district/town alternative; do not retain coordinates.                                    |
| Language, literacy, and accessibility   | English-only or text-heavy interaction can exclude the intended community.                                                            | Bengali-first labels, smaller English support text, large targets, familiar wording, optional voice, and plain-language testing.     |
| Equity and bias                         | The product may work differently across age, gender, disability, literacy, device access, and districts.                              | Test with diverse rural users/caregivers; offer human support; avoid making AI or smartphone access a prerequisite for guidance.     |
| Service-data accuracy                   | Outdated clinic/camp/phone information may cause unnecessary travel or delay.                                                         | Show verified-at/demo status; assign outreach ownership; define update cadence and escalation for stale entries.                     |
| Human escalation and autonomy           | Users should not be forced to rely solely on automation.                                                                              | Provide ASHA, PHC, clinic, family-sharing, restart, and service-navigation options in every appropriate branch.                      |
| Transparency and feedback               | Users and stakeholders need to understand limitations and report problems.                                                            | Explain purpose and privacy in simple language; offer feedback/help path; maintain non-identifying audit events and review process.  |

### 6.1 Governance and release gate

- Clinical/content reviewer approves rules, urgent copy, Bengali scripts, and changes affecting safety.
- Outreach coordinator verifies service/camp details and owns freshness updates.
- Product owner approves scope and evidence claims; engineering owner controls feature flags and deployment.
- Privacy/security review confirms no PII/secrets in logs, browser bundle, analytics, or Git.
- Pilot release requires tested fallback for AI, location, voice, and service lookup failures.

## 7. Submission checklist

| Item                 | Submission-ready evidence                                                                       |
|----------------------|-------------------------------------------------------------------------------------------------|
| Lean Canvas          | Completed Section 2, validated assumptions clearly marked.                                      |
| Methodology          | Section 3 explains Lean UX + Agile Kanban and delivery cadence.                                 |
| Project plan         | Section 4 with actual dates/owners substituted if required.                                     |
| Jira board           | Real screenshot inserted in Section 5.1 after board capture.                                    |
| Ethics               | Section 6 safeguards linked to the PRD/HLD and live product behavior.                           |
| Supporting artefacts | PRD, HLD, wireframes, Jira board/dashboard, and GitHub repository links included in final pack. |